

# L'Hôpital's Rule: Statement and Proof

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## 1 Prerequisite

Rolle's theorem is taken as known and is not reproved here.

**Theorem 1.1** (Rolle). *Let  $h : [p, q] \rightarrow \mathbb{R}$  be continuous on  $[p, q]$  and differentiable on  $(p, q)$ , with  $h(p) = h(q)$ . Then there exists  $\xi \in (p, q)$  such that  $h'(\xi) = 0$ .*

## 2 Cauchy's Mean Value Theorem

**Lemma 2.1** (Cauchy). *Let  $f, g : [p, q] \rightarrow \mathbb{R}$  be continuous on  $[p, q]$  and differentiable on  $(p, q)$ . Then there exists  $\xi \in (p, q)$  such that*

$$[f(q) - f(p)] g'(\xi) = [g(q) - g(p)] f'(\xi).$$

*If in addition  $g'(x) \neq 0$  for every  $x \in (p, q)$ , then  $g(p) \neq g(q)$  and*

$$\frac{f(q) - f(p)}{g(q) - g(p)} = \frac{f'(\xi)}{g'(\xi)}.$$

*Proof.* Define  $h : [p, q] \rightarrow \mathbb{R}$  by

$$h(x) = [f(q) - f(p)] g(x) - [g(q) - g(p)] f(x).$$

As a linear combination of  $f$  and  $g$ ,  $h$  is continuous on  $[p, q]$  and differentiable on  $(p, q)$ . A direct computation gives

$$h(p) = f(q)g(p) - g(q)f(p), \quad h(q) = f(q)g(p) - g(q)f(p),$$

so  $h(p) = h(q)$ . By Rolle's theorem there is  $\xi \in (p, q)$  with  $h'(\xi) = 0$ ; since

$$h'(x) = [f(q) - f(p)] g'(x) - [g(q) - g(p)] f'(x),$$

this is exactly the claimed identity.

Now suppose  $g'(x) \neq 0$  throughout  $(p, q)$ . If  $g(p) = g(q)$ , Rolle's theorem applied to  $g$  itself on  $[p, q]$  would produce a point of  $(p, q)$  at which  $g'$  vanishes, contradicting the hypothesis. Hence  $g(p) \neq g(q)$ , and since also  $g'(\xi) \neq 0$  (else the identity above would force  $g(p) = g(q)$  as well, again a contradiction), dividing by  $g'(\xi)[g(q) - g(p)]$  yields the second formula.  $\square$

### 3 L'Hôpital's Rule

**Theorem 3.1** (L'Hôpital). *Let  $-\infty \leq c < d \leq +\infty$ , and let  $f, g : (c, d) \rightarrow \mathbb{R}$  be differentiable, with  $g'(x) \neq 0$  for every  $x \in (c, d)$ . Suppose*

$$\lim_{x \rightarrow c^+} \frac{f'(x)}{g'(x)} = A, \quad A \in \mathbb{R} \cup \{-\infty, +\infty\},$$

(when  $c = -\infty$ , “ $x \rightarrow c^+$ ” is read simply as  $x \rightarrow -\infty$ ), and suppose in addition that either

- (a)  $\lim_{x \rightarrow c^+} f(x) = \lim_{x \rightarrow c^+} g(x) = 0$ , or  
 (b)  $\lim_{x \rightarrow c^+} g(x) = +\infty$  (or  $-\infty$ ).

Then

$$\lim_{x \rightarrow c^+} \frac{f(x)}{g(x)} = A.$$

*Proof. Step 0:  $g$  is injective on  $(c, d)$ .* If  $g(x) = g(y)$  for some  $x \neq y$  in  $(c, d)$ , Rolle's theorem applied to  $g$  on the interval between  $x$  and  $y$  would produce a zero of  $g'$  in  $(c, d)$ , contradicting the hypothesis  $g' \neq 0$ . So  $g$  takes each value at most once on  $(c, d)$ ; being also continuous, it is strictly monotonic there (a standard consequence of the intermediate value theorem).

**Step 1: the case  $A \in \mathbb{R}$ .** Fix  $\varepsilon > 0$  and choose real numbers  $q, r$  with

$$A - \varepsilon < q < A < r < A + \varepsilon.$$

Since  $f'/g' \rightarrow A$  as  $x \rightarrow c^+$ , there is a point  $b \in (c, d)$  such that

$$q < \frac{f'(t)}{g'(t)} < r \quad \text{for all } t \in (c, b). \quad (1)$$

For any  $x, y$  with  $c < x < y < b$ , Step 0 gives  $g(x) \neq g(y)$ , so Cauchy's theorem (Lemma 1) applies on  $[x, y]$ : there exists  $\xi = \xi(x, y) \in (x, y) \subset (c, b)$  with

$$\frac{f(x) - f(y)}{g(x) - g(y)} = \frac{f'(\xi)}{g'(\xi)}.$$

Combined with (1), this gives

$$q < \frac{f(x) - f(y)}{g(x) - g(y)} < r \quad \text{for all } c < x < y < b. \quad (2)$$

*Case (a).* Fix  $y \in (c, b)$ . By Step 0,  $g$  is strictly monotonic on  $(c, d)$  with  $\lim_{x \rightarrow c^+} g(x) = 0$ , so 0 is not attained on  $(c, d)$ : hence  $g(y) \neq 0$ . Letting  $x \rightarrow c^+$  in (2), and using  $f(x) \rightarrow 0$ ,  $g(x) \rightarrow 0$ , the middle term tends to  $f(y)/g(y)$ , and weak inequalities survive the limit:

$$q \leq \frac{f(y)}{g(y)} \leq r \quad \text{for every } y \in (c, b). \quad (3)$$

*Case (b).* Fix  $y \in (c, b)$ . Because  $g(x) \rightarrow +\infty$  as  $x \rightarrow c^+$  (the case  $g \rightarrow -\infty$  is identical after replacing  $g$  by  $-g$ ), we may choose  $x$  close enough to  $c$  that  $g(x) > 0$  and  $g(x) > g(y)$ . Multiplying (2) by the positive quantity  $[g(x) - g(y)]/g(x)$  gives

$$q \left(1 - \frac{g(y)}{g(x)}\right) < \frac{f(x)}{g(x)} - \frac{f(y)}{g(x)} < r \left(1 - \frac{g(y)}{g(x)}\right). \quad (4)$$

Let  $x \rightarrow c^+$  with  $y$  fixed. Since  $g(x) \rightarrow +\infty$ , both  $g(y)/g(x) \rightarrow 0$  and  $f(y)/g(x) \rightarrow 0$ , so (4) yields

$$q \leq \liminf_{x \rightarrow c^+} \frac{f(x)}{g(x)} \leq \limsup_{x \rightarrow c^+} \frac{f(x)}{g(x)} \leq r. \quad (5)$$

In either case, (3) or (5) gives

$$q \leq \liminf_{x \rightarrow c^+} \frac{f(x)}{g(x)} \leq \limsup_{x \rightarrow c^+} \frac{f(x)}{g(x)} \leq r.$$

This holds for *every* admissible pair  $q, r$  with  $A - \varepsilon < q < A < r < A + \varepsilon$ . Letting  $\varepsilon \rightarrow 0^+$  forces  $q \rightarrow A$ ,  $r \rightarrow A$ , hence

$$\liminf_{x \rightarrow c^+} \frac{f(x)}{g(x)} = \limsup_{x \rightarrow c^+} \frac{f(x)}{g(x)} = A,$$

i.e.,  $\lim_{x \rightarrow c^+} f(x)/g(x) = A$ .

**Step 2: the case  $A = +\infty$  (the case  $A = -\infty$  is symmetric).** Fix any  $r \in \mathbb{R}$ . Since  $f'/g' \rightarrow +\infty$ , there is  $b \in (c, d)$  with  $f'(t)/g'(t) > r$  for all  $t \in (c, b)$ . Repeating the argument of Step 1 using only this one-sided bound (there is no companion  $q$  to track) produces, in either case (a) or (b),

$$\liminf_{x \rightarrow c^+} \frac{f(x)}{g(x)} \geq r.$$

Since  $r \in \mathbb{R}$  was arbitrary,  $\liminf_{x \rightarrow c^+} f(x)/g(x) = +\infty$ , i.e.,  $\lim_{x \rightarrow c^+} f(x)/g(x) = +\infty$ .  $\square$

## 4 Extensions

**Corollary 4.1** (Left-hand and two-sided limits). *The theorem holds equally for  $x \rightarrow c^-$ : if  $f, g$  are differentiable on  $(e, c)$  for some  $e < c$ , with  $g' \neq 0$  there, and  $\lim_{x \rightarrow c^-} f'(x)/g'(x) = A$  together with the analogous hypothesis (a) or (b), then  $\lim_{x \rightarrow c^-} f(x)/g(x) = A$ . Indeed, set  $\tilde{f}(x) = f(2c - x)$  and  $\tilde{g}(x) = g(2c - x)$  on  $(c, 2c - e)$ ; as  $x \rightarrow c^+$ ,  $2c - x \rightarrow c^-$ , and  $\tilde{f}'(x)/\tilde{g}'(x) = f'(2c - x)/g'(2c - x) \rightarrow A$ , so Theorem 2 applies to  $\tilde{f}, \tilde{g}$  and yields the claim after undoing the substitution. Consequently, if  $f, g$  are differentiable on a full punctured neighborhood  $(e, c) \cup (c, d)$  of a finite point  $c$  and both one-sided derivative-ratio limits equal  $A$ , then  $\lim_{x \rightarrow c} f(x)/g(x) = A$ .*

**Corollary 4.2** (Limits at infinity). *The case  $x \rightarrow +\infty$  reduces to Theorem 2 via  $t = 1/x$ : if  $F(t) = f(1/t)$ ,  $G(t) = g(1/t)$  for  $t$  near  $0^+$ , the chain rule gives*

$$\frac{F'(t)}{G'(t)} = \frac{f'(1/t)}{g'(1/t)},$$

*so  $\lim_{x \rightarrow +\infty} f'(x)/g'(x) = A$  implies  $\lim_{t \rightarrow 0^+} F'(t)/G'(t) = A$ ; hypotheses (a)/(b) transfer identically, so Theorem 2 gives  $\lim_{t \rightarrow 0^+} F(t)/G(t) = A$ , which is  $\lim_{x \rightarrow +\infty} f(x)/g(x) = A$ . The case  $x \rightarrow -\infty$  is analogous via  $t = -1/x$ .*

**Remark 4.3.** The hypothesis that  $\lim_{x \rightarrow c^+} f'(x)/g'(x)$  exists cannot be dropped, and the theorem is not reversible. Take  $f(x) = x + \sin x$ ,  $g(x) = x$  on  $(0, \infty)$ ; both diverge to  $+\infty$  as  $x \rightarrow \infty$ , and

$$\frac{f(x)}{g(x)} = 1 + \frac{\sin x}{x} \rightarrow 1,$$

so the quotient converges. Yet

$$\frac{f'(x)}{g'(x)} = 1 + \cos x$$

oscillates between 0 and 2 and has no limit as  $x \rightarrow \infty$ . This is not a contradiction: the theorem states a *sufficient*, not necessary, condition for evaluating  $\lim f/g$ .